

Question:

Darrell and Allen's ages are in the ratio of 7:11. If their total age now is 162, calculate Allen's age 10 years from now.

Reasoning + Answer:

<think> To find Allen's age 10 years from now, we need to first determine the current ages of Darrell and Allen. Given that their ages are in the ratio 7 : 11, let's assume Darrell's age is $7x$ and Allen's age is $11x$. The total of their ages is given as 162, so we can set up the equation $7x + 11x = 162$. Combining like terms, we get $18x = 162$. Dividing both sides by 18, we find $x = 9$. Now that we know x , we can calculate Darrell's and Allen's current ages: Darrell's age is $7x = 7 * 9 = 63$, and Allen's age is $11x = 11 * 9 = 99$. To find Allen's age 10 years from now, we add 10 to his current age: $99 + 10 = 109$. </think><answer> 109</answer>

